

TENOF 300MG TABLET

Name of the product

Tenof 300mg Tablet

Name and strength of Active Ingredient

Name : Tenofovir Disoproxil Fumarate

Strength : 300mg

300 mg of Tenofovir disoproxil fumarate equivalent to 245 mg of Tenofovir disoproxil.

Product Description

Light blue coloured, almond shaped, film coated tablets, debossed with 'H' on one side and '123' on other side.

Pharmacodynamics/Pharmacokinetics

Tenofovir DF is hydrolysed to tenofovir, which is then phosphorylated by cellular kinases to the pharmacologically active metabolite tenofovir diphosphate. Tenofovir diphosphate inhibits the activity of HIV reverse transcriptase by competing with the nucleotide deoxyadenosine 5'-triphosphate for incorporation into viral DNA. Once incorporated into viral DNA, it terminates DNA elongation because of lack of a ribose ring.

Pharmacokinetics

Absorption and Distribution

The oral bioavailability of tenofovir after administration of tenofovir DF 300 mg/day was 25% and increased to 39% when tenofovir DF was administered with a standardized high fat meal. Median steady-state maximum serum tenofovir concentrations (C_{max}) and the area under the serum tenofovir concentration-time curve (AUC) were 326 ng/mL and 3020 ng • h/mL in patients infected with HIV who received tenofovir DF 300 mg/day with food for 28 days. The median time to C_{max} was 2.3 hours.

Metabolism and Elimination

Tenofovir concentrations in serum decline in a biphasic manner.

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Indications

HIV-1 Infection

Tenofovir is indicated in combination with other antiretroviral agents for the treatment of HIV-1 infection in adults and pediatric patients 12 years of age and older.

The following points should be considered when initiating therapy with tenofovir for the treatment of HIV-1 infection:

- TENOF should not be used in combination with other products containing tenofovir disoproxil fumarate.

Chronic Hepatitis B

Tenofovir is indicated for the treatment of chronic hepatitis B in adults.

The following points should be considered when initiating therapy with tenofovir for the treatment of HBV infection:

- This indication is based primarily on data from treatment of subjects who were nucleoside-treatment-naïve and a smaller number of subjects who had previously received lamivudine or adefovir dipivoxil. Subjects were adults with HBeAg-positive and HBeAg-negative chronic hepatitis B with compensated liver disease.
- Tenofovir was evaluated in a limited number of subjects with chronic hepatitis B and decompensated liver disease.
- The numbers of subjects in clinical trials who had lamivudine- or adefovir associated substitutions at baseline were too small to reach conclusions of efficacy.

Recommended dose

For the treatment of HIV-1 or chronic hepatitis B: The dose of Tenofovir is 300 mg once daily taken orally, without regard to food.

In the treatment of chronic hepatitis B, the optimal duration of treatment is unknown.

Dose Adjustment for Renal Impairment in Adults

Significantly increased drug exposures occurred when Tenofovir was administered to subjects with moderate to severe renal impairment. Therefore, the dosing interval of Tenofovir should be adjusted in patients with baseline creatinine clearance <50 mL/min using the recommendations in

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Table 1. These dosing interval recommendations are based on modelling of single-dose pharmacokinetic data in non-HIV and non-HBV infected subjects with varying degrees of renal impairment, including end-stage renal disease requiring haemodialysis. The safety and effectiveness of these dosing interval adjustment recommendations have not been clinically evaluated in patients with moderate or severe renal impairment, therefore clinical response to treatment and renal function should be closely monitored in these patients.

No dose adjustment is necessary for patients with mild renal impairment (creatinine clearance 50–80 mL/min). Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment.

Table 1 Dosage Adjustment for Patients with Altered Creatinine Clearance

	Creatinine Clearance (mL/min) ^a			Haemodialysis Patients
	>50	30-49	10-29	
Recommended 300 mg Dosing Interval	Every 24 hours	Every 48 hours	Every 72 to 96 hours	Every 7 days or after a total of approximately 12 hours of dialysis ^b

a. Calculated using ideal (lean) body weight.

b. Generally once weekly assuming three haemodialysis sessions a week of approximately 4 hour duration.

Tenofovir should be administered following completion of dialysis.

The pharmacokinetics of tenofovir have not been evaluated in non-haemodialysis patients with creatinine clearance <10 mL/min; therefore, no dosing recommendation is available for these patients.

- No data are available to make dose recommendations in paediatric patients 12 years of age and older with renal impairment.

Mode of administration

Oral route of administration

Contraindications

None

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Warnings and precautions**Lactic Acidosis/Severe Hepatomegaly with Steatosis**

Lactic acidosis and severe hepatomegaly with steatosis (sometimes fatal) have been reported rarely in patients receiving nucleoside reverse transcriptase inhibitors (NRTIs) alone or in conjunction with other antiretroviral agents.

Caution should be observed when nucleoside analogs are used in patients with known risk factors for liver disease; however, lactic acidosis and severe hepatomegaly with steatosis have been reported in patients with no known risk factors.

Patients Co-infected with HIV and HBV

The need to avoid antiviral resistance complicates the selection of active agents. Resistance to HIV therapy limits the choices for treatment of HBV infection. Tenofovir may select for resistance mutations in HIV polymerase and hence tenofovir should only be used in HIV-1 and HBV coinfecting patients as part of an appropriate antiretroviral combination regimen.

HIV-1 antibody testing should be offered to all HBV-infected patients before initiating therapy with tenofovir. It is also recommended that all patients with HIV-1 be tested for the presence of chronic hepatitis B before initiating treatment with tenofovir.

Renal Impairment

Although tenofovir did not show any significant cytotoxicity in isolated human RPTECs in an in vitro study, tenofovir has been associated with changes in renal function in vivo. Changes in renal function (incidence of hypophosphataemia and elevations in serum creatinine) in antiretroviral-naïve patients were similar for those receiving tenofovir DF- or stavudine-based therapy after 96 weeks. However, rare episodes of acute renal failure have been observed in patients receiving tenofovir DF in combination with other with antiretroviral drugs.

Additional safety recommendations include the following: i) the dosing interval of tenofovir DF should be adjusted in patients with baseline Cl_{cr} , < 50 ml/min; ii) tenofovir DF should be avoided with concurrent or recent use of nephrotoxic agents; and iii) patients at risk for, or with a history

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of renal dysfunction, and patients receiving concomitant nephrotoxic agents should be carefully monitored for changes in Cr and phosphorus.

Decreases in Bone Mineral Density

Bone monitoring should be considered for HIV-infected patients who have a history of pathologic bone fracture or are at substantial risk for osteopenia.

Fat Redistribution

In HIV-infected patients, redistribution/accumulation of body fat including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement, and "cushingoid appearance" have been observed in patients receiving combination antiretroviral therapy.

Immune Reconstitution Syndrome

Immune reconstitution syndrome has been reported in patients treated with combination antiretroviral therapy, including tenofovir disoproxil fumarate.

Early Virologic Failure

Patients on a therapy utilizing a triple nucleoside-only regimen should be carefully monitored and considered for treatment modification.

Effect on ability to drive and use machines

This medicine may affect your ability to drive or use machines. If the tablets make you feel sick, dizzy or tired, or give you a headache, do not drive or use machines and contact your doctor immediately

Interaction with other Medicaments

Notable drug interactions occur when tenofovir DF is administered with didanosine or atazanavir. Otherwise, tenofovir DF can be administered with all other antiretrovirals without dose adjustments. Furthermore, there are no relevant pharmacokinetic interactions with methadone, oral contraceptives, ribavirin, or rifampacin.

Tenofovir DF and didanosine

Administration of tenofovir DF with ddI is not recommended.

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Tenofovir DF and atazanavir

The current recommendation for administration of atazanavir with tenofovir is to use atazanavir 300 mg and ritonavir 100 mg, both once daily. The administration of atazanavir/ritonavir with tenofovir resulted in increased exposure to tenofovir by C_{min} , C_{ss} , and AUC. Higher tenofovir concentrations could potentiate tenofovir-associated adverse events, including renal disorders.

Nucleotide combinations

Avoid the unfavourable nucleoside combinations of tenofovir DF plus abacavir or ddI for initial antiretroviral therapy.

Patients with Impaired Renal Function

It is recommended that the dosing interval for tenofovir DF be modified in patients with creatinine clearance <50 mL/min or in patients with ESRD who require dialysis.

Pregnancy and lactation

Pregnancy

Pregnancy Category B

Tenofovir should be used during pregnancy only if clearly needed.

Lactation

The Centers for Disease Control and Prevention recommend that women infected with HIV not breastfeed their infants in order to avoid postnatal spread of the virus through breast milk.

Pediatric Use

Safety and efficacy in pediatric patients have not been established and there is no recommended dose for children.

Side effects

Tenofovir DF has a limited effect on mitochondrial DNA polymerase.

The most common treatment-related adverse events are predominantly gastrointestinal nature and include: diarrhoea, headache, vomiting, flatulence, abdominal pain and anorexia.

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Symptoms and treatment of overdose

If overdose occurs the patient must be monitored for evidence of toxicity, and standard supportive treatment applied as necessary. Tenofovir is efficiently removed by hemodialysis with an extraction coefficient of approximately 54%. Following a single 300 mg dose of tenofovir disoproxil fumarate, a 4 -hour hemodialysis session removed approximately 10% of the administered tenofovir dose.

Storage condition

Store below 30⁰C and protect from moisture.

Shelf life: 2 Years

Dosage forms and packaging available

30 Tablets in HDPE container

Manufacturer

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